

2022

COMPUTER SCIENCE — GENERAL

Paper : SEC-A-X-1

(Communication, Computer Network and Internet)

Full Marks : 80

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question nos. 1 and 2** and **any four** questions from the rest.

1. Answer **any ten** questions :

2×10

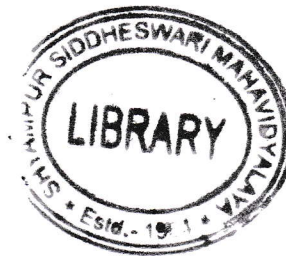
- (a) What is firewall?
- (b) What are cookies?
- (c) How does guided media differ from unguided media?
- (d) What is the functionality of ISDN?
- (e) What is the purpose of transport layer in TCP/IP model?
- (f) Mention the function of hub.
- (g) What are the types of LAN technologies?
- (h) Find out the address class of the given IP address : 191.254.200.4
- (i) Write the advantages of co-axial communication media over twisted pair.
- (j) What do you mean by multipoint communication?
- (k) Differentiate between logical address and physical address of a device.
- (l) State some functions of a web browser.
- (m) Define a gateway.
- (n) What are the types of noise that can affect data communication?
- (o) Differentiate between periodic and non-periodic signal.

2. Write short notes on **any four** of the following :

5×4

- (a) Ring Topology
- (b) Router
- (c) www
- (d) Video Conferencing

Please Turn Over



- (e) Satellite communication
- (f) PSK.
3. (a) Compare LAN and WAN.
- (b) What do you mean by attenuation? Mention a possible remedy for attenuation.
- (c) What is a bridge? 5+(2+1)+2
4. (a) Compare ASK and FSK.
- (b) Define QAM.
- (c) What is the relationship between bit rate and baud rate? 5+3+2
5. (a) What is the difference between analog and digital transmission? Explain it.
- (b) What are the roles of data link layer and network layer in OSI model? Explain.
- (c) What is ARPANET? 4+4+2
6. (a) Compare FDM and TDM.
- (b) Differentiate between NRZ and RZ line coding with examples.
- (c) Define bandwidth of an analog signal. 5+3+2
7. (a) What is the advantage of using Manchester line coding?
- (b) Write a short note on ADSL.
- (c) Compare UDP and TCP protocols. 2+5+3
8. (a) Discuss about any two protocols needed for sending and receiving E-mail.
- (b) Discuss FQDN and PQDN in the context of DNS. Give proper examples. 5+5
9. (a) Explain the concept and steps of Pulse Code Modulation (PCM) in details.
- (b) Differentiate DM and PCM.
- (c) Define signal to noise ratio. 5+3+2
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